

Mining Egocentric Video with Sorter Speech for Plastic Classification

Yeonu Lee

Mountain View High School
High School Student
Mountain View, CA, USA

Jamalia Sultana

Dept. of Computer Science
Stony Brook University
Stony Brook, NY, USA

Zhaozheng Yin

Dept. of Computer Science
Stony Brook University
Stony Brook, NY, USA

Abstract—Sorting household plastics by hand remains standard practice at material recovery facilities (MRF), where it poses health issues for workers and problem of insufficient throughput to meet recycling demand. In order to address this problem, we mined Egowaste – a dataset produced by material recovery facilities, where they paired visual and spoken records of the items workers handle. Specifically, it is a first-person dataset in which each clip captures a worker holding a plastic item while describing it aloud. Our pipeline first masks the target object using a segmentation model, then selects the best candidate mask through a multi-stage scoring process: contrastive and supervised image models score each candidate crop, while a audio branch transcribes the sorter’s speech using a LoRA-adapted Whisper model and matches the transcript against a phrase list to identify the spoken plastic category. Image-based and speech-based predictions are then combined via logistic regression to produce a final classification. We evaluate the system at multiple stages, from mask overlap (intersection-over-union, IoU) to clip-level “any frame correct” accuracy. The best-performing vision-speech fusion setup achieves a group-level any-frame accuracy of 73.4%, compared with 69.6% for the strongest vision-only model. We present stage-by-stage performance scores and confusion matrices for the mask, crop, and fusion configurations.

Index Terms—multimodal data mining, recycling and waste management, plastic classification, first-person vision, multi-modal fusion

I. INTRODUCTION

Recycling facilities generate material-wise waste sorting data, including video and the verbal descriptions sorters give as they work. Every item that passes a sorting station leaves a frame containing the image of the hand and the object, along with a spoken phrase that names the material or describes its characteristics, which is associated with a plastic class label. Mining these paired records means learning which appearances and which phrases go with each plastic type, which is exactly the decision a sorter makes during the waste sorting process. We therefore study how to mine this multimodal data to support fine-grained plastic sorting, a key step in waste management. Automated waste-sorting shows up in many places: smart bins, conveyor belts in MRFs, outdoor litter cameras, underwater cleanup, and construction sites [1], [2]. Most existing work relies solely on RGB imagery, and while surveys report strong performance on clean lab sets, these models still struggle with overlapping, deformed, or visually similar plastic objects [3]–[5]. Even setups that add sensing beyond RGB [6], [7] watch the stream from a fixed camera. None

of them leverage the *sorter’s first-person viewpoint* or the short verbal phrases workers use to distinguish visually similar plastics. A fully automatic line is also hard to run in practice. In several regions, MRF rules require a certified operator or supervisor on site, people checking the belt, and a plan for hazardous items [8]–[11]. Thus, we suggest the vision being a *helper for the sorter*, not a replacement. We target MRF-style detailed plastic sorting while keeping the human sorter in the process. Prior first-person datasets such as Ego4D [12] and IndEgo [13] (covering factory assembly, logistics, and spoken narration) already study egocentric hand-object interaction, but neither addresses MRF plastic categories or descriptive audio. This setting is challenging as target objects are small, the hand occludes part of the object, and spoken phrases do not map one-to-one onto class labels. Table I shows that existing collections do not capture this gap.

This paper makes four contributions:

- We evaluate end-to-end plastic sorting using the EgoWaste camera-1 subset, which contains **15 categories** and 463 video clips (3,538 frames).
- We use **SAM3** [14] to cut out the object, then a **four-stage mask pick and rerank** so later crops are cleaner.
- We combine **image models** (CLIP [15], SigLIP [16], EfficientNet-B0 [17]) with **Whisper LoRA** transcription [18], [19] and **phrase matching** (Sentence Transformer MiniLM-L6-v2 [20]) on our common-phrase list, then join the two with logistic regression.
- We report **stage-by-stage scores** from mask IoU to frame- and clip-level Top-1 accuracy, to assess the contributions of cropping, reranking, and speech at each stage of the pipeline.

Section II reviews related work. Section III describes the dataset and task. Section IV presents our pipeline. Section V reports experiments; Sections VI–VII discuss limits and next steps.

II. RELATED WORK

Public waste datasets differ in scene, view, audio, and label granularity (Table I). TrashNet [3] is a common clean benchmark: one object on a plain background, with little overlap, motion, or factory lighting [2]. Light CNNs and mixed designs (EfficientNet, MobileNet, CustomNet) trained on TrashNet and merged household photos report >94–98% accuracy on

TABLE I
COMPARISON OF WASTE DATASETS VS. EGOWASTE CAMERA-1 PLASTICS
(THIS WORK). SCENARIO TAGS FOLLOW [2].

Dataset	Scenario	View	Audio	Classes
TrashNet	smart bin	studio	no	6
TACO	outdoor	in-the-wild	no	60
ZeroWaste	MRF belt	conveyor	no	4
WaRP	MRF belt	conveyor	no	28
UrbanWaste	smart bin	in-bin	no	193
EgoWaste	MRF (ours)	first-person	yes	15

single, centered objects on Jetson or Raspberry Pi [2], [3]. TACO [21] labels outdoor litter under a 28-class tree with COCO-style instance masks, with class frequencies so uneven that papers often evaluate on merged groups. Litter systems fine-tune Mask R-CNN or YOLO on TACO; underwater sets such as TrashCan and construction-waste detectors use different physics and class lists [2], [21]. Plant-scale methods find flexible conveyor streams with YOLO detectors, hierarchical heads, or weakly labeled segmentation: ZeroWaste [4] has semi- and weakly labeled variants, while WaRP [5] first finds five broad groups, then fine classes (mAP_{50} 58.6%), with overlapping flexible objects and masks trained from class labels only. UrbanWaste [22] watches household bins with before/after change detection (13,553 pairs). Open-vocabulary segmentation (SAM3 [14], Grounding DINO [23]) and vision-language models (CLIP/SigLIP [15], [16]) cut the cost of training a new detector for each plastic type when labeled data are scarce. First-person fusion more often targets action recognition [12], [24], [25] than material type. Plants that cannot separate black film or mixed materials from RGB alone add NIR or induction sensing on a fixed camera [6], [7].

EgoWaste instead records pixel-level masks over 463 video clips and pairs each clip with the verbal phrases sorters use while working. The extra modality we exploit is therefore the worker’s own speech, not the impact sounds [7], [26] or spectral readings earlier multimodal sorters rely on.

III. DATASET AND TASK FORMULATION

We use the EgoWaste camera-1 plastics shareable evaluation subset. Each record has an RGB frame, a pixel-level true mask, a 15-way class label, a bounding box, and a matching video clip at the toss. The evaluation split has **3,538** frames across **463** clips and **15** plastic types in a resin-based class list. Classes include deposit and non-deposit bottles, numbered containers (#1 clamshells, #2 natural/colored, #3–7), black plastic, garbage bags, and leftover groups such as unnumbered fragments. See Table I and Sec. II for how this subset differs from prior waste collections.

In each video clip, workers say short phrases while they sort (for example, “number one clamshell,” “five PP”). We extract the soundtrack from each video clip and transcribe it. We keep 104 unique *common phrases* as training metadata and as a search index from Whisper text to class labels. Fig. 1 shows a frame–mask pair and class crops; spoken phrases are listed in the caption.



(a) Frame + GT mask (GX010023, Black Plastic).



(b) category_name + GT masked crop.

Fig. 1. Dataset samples. Spoken common phrases in (b), left to right, top to bottom: “unnumbered black plastic,” “one clamshell,” “two colors,” “natural,” “three to seven,” “five,” “one deposit,” “non-deposit garbage bag,” “other non-deposit,” “non-deposit one,” “film plastic,” “do we have other plastic?,” “bulky rigid,” “unnumbered,” “other plastic.”

IV. METHOD

Fig. 2 shows the pipeline. On each frame, SAM3 [14] proposes object masks, we classify the crops, and we apply margin rerank by the gap between the top two class scores. On each clip we extract the soundtrack, transcribe it with Whisper+LoRA, and map it to a class with MiniLM phrase matching and text rules; an agreement bonus from speech to vision can boost a matching crop, and late fusion logistic regression joins the two branches.

A. Problem Setup

We score at the **frame level** (each labeled frame) and the **group level** (463 video clips). A clip is correct under an *any-frame any-bbox* rule: if any frame’s predicted class matches the clip label, using the best mask for that setting, the clip counts. The main end-to-end number is **final-probability Top-1** after we drop failed masks and calibrate probabilities. We also report mask IoU versus the true outline and segment-level class Top-1 accuracy in Table II.

B. Detection and Mask Selection

Hand-crop detection asks SAM3 [14] only inside a hand-guided region and keeps the in-hand mask that sits most inside that region, or the largest box if that fails. **Full-image detection** runs Top-5 plastic segmentation over the whole frame. Mask IoU rises from 8.3% (hand crop) to 19.8% (full image). Tight hand crops often cut off the object.

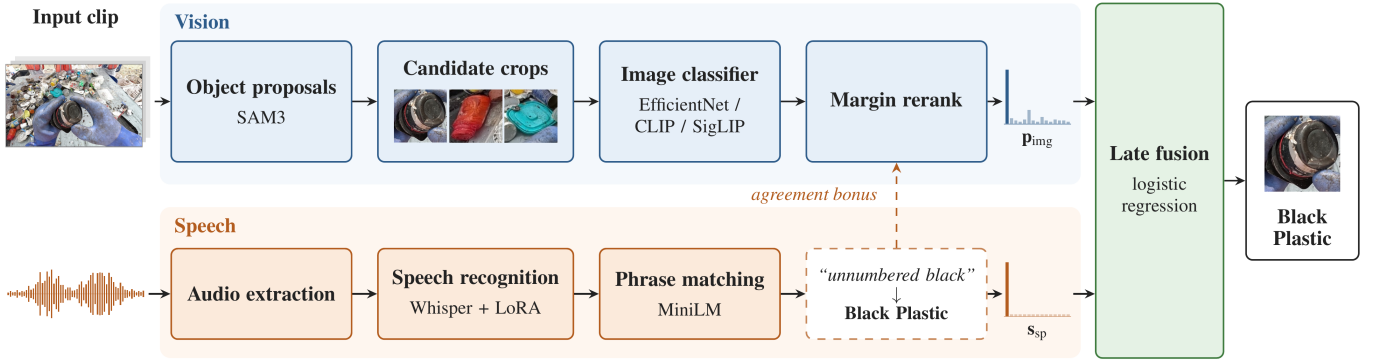


Fig. 2. System overview for one input clip. **Vision branch** (per frame): SAM3 object proposals $\rightarrow$ candidate crops $\rightarrow$ image classifier (EfficientNet-B0, CLIP, or SigLIP) $\rightarrow$ margin rerank by the gap between the top two class scores, yielding the 15-D score vector $\mathbf{p}_{\text{img}}$. **Speech branch** (per clip): audio extraction $\rightarrow$ Whisper+LoRA speech recognition $\rightarrow$ MiniLM phrase matching, which maps the spoken phrase (here “unnumbered black”) to a class and yields the one-hot vector $\mathbf{s}_{\text{sp}}$. The dashed *agreement bonus* feeds that spoken class back into margin rerank. Late fusion logistic regression joins $\mathbf{p}_{\text{img}}$ and $\mathbf{s}_{\text{sp}}$ into the final class (here Black Plastic).

Phases 1–4 then clean up the mask list: Ph.1 classifies crops from the two detection strategies; Ph.2 adds candidates, optional masked crops, and padded boxes (the main accuracy jump is picking a better mask); Ph.3 uses Top-5 pools that mix in-hand, full-image, and score-ranked masks; Ph.4 reranks by score margin and, at clip level, gives extra credit when Whisper matches vision. For each frame we keep up to five mask candidates. We score them by how much they sit in the hand region, how they overlap SAM3 [14] confidence, and simple edge/contact checks from the segment code. Reranking keeps the candidate with the largest gap between the top two classifier scores, and may add a bonus when Whisper agrees with vision. Masked crops black out pixels outside the mask before resize. Box crops with 15% padding are a lighter backup when the mask edge is shaky.

C. Vision Classifiers

We train all image models on the same train/validation split, sampling so each class appears in proportion to the data. CLIP and SigLIP linear probes train one layer on frozen patch features. LoRA adapters sit on attention projections with rank 8. EfficientNet-B0 is fine-tuned end-to-end with standard augmentations (random resized crop, color jitter). We pick checkpoints on validation final-probability Top-1 and run them once on held-out evaluation frames. Crops (box by default; masked crops when we turn that piece off) go into:

- **Zero-shot:** CLIP and SigLIP, each with 15-class prompts.
- **Linear probe & LoRA** for CLIP ViT-B/32 and SigLIP.
- **EfficientNet-B0:** supervised fine-tuning on training folds.

Classifiers output 15-way class scores that sum to 1.

D. Audio Pipeline

We extract audio from each video clip and transcribe it with **Whisper large-v2 + LoRA** (rank 8, $\alpha=32$ on query/value projections). Transcripts map to classes in two ways: (i) **meaning-based search:** embed the transcript and all common phrases with a MiniLM sentence encoder [20], take the closest phrase,

then read its class; (ii) **simple text rules** for common speech-to-text errors (for example, “non deposit garbage bag” $\rightarrow$ Non-Deposit). We cache per-clip phrase maps for fusion training and test.

E. Late Fusion and Metrics

At the end we join the 15-D image scores with a 15-D one-hot speech class (zeros if unknown) and train **logistic regression** on the training split. Two fusion variants pair CLIP or EfficientNet images with Whisper text (*Fusion*, *Fusion-EN*). Metrics: mask IoU, segment class Top-1, frame/group final-probability Top-1, and how often the pipeline succeeds.

V. EXPERIMENTS AND RESULTS

All experiments run in PyTorch on NVIDIA GPUs. SAM3 [14] produces hand-crop and full-image detections; exported masks feed mask selection and reranking in Table II. Whisper LoRA uses learning rate 10^{-4} , batch 8, and 3 epochs on toss-audio transcripts. Phrase search uses 384-D MiniLM embeddings; cosine similarity picks the Top-1 common phrase, then we look up the class. Fusion logistic regression trains on joined 30-D training-split features. Evaluation rolls per-frame JSON into clip scores by majority vote and any-frame any-bbox. A fixed scoring script compares five classifier families—CLIP and SigLIP linear probes, EfficientNet-B0, Fusion (CLIP+Whisper), and Fusion-EN (EfficientNet+Whisper)—at the representative settings of each phase in Table II. On the final pipeline, EfficientNet segment class Top-1 is 61.2% on 3,538 frames; final-probability Top-1 (67.8%) uses the same mask filter as our reported pipeline.

A. Phase-Wise End-to-End Accuracy

Table II lists frame-level final-probability Top-1 (%) by stage and a group row over 463 clips (any-frame any-bbox). Raising mask IoU from 8.3% (hand crop) to 19.8% (full-image detection) helps, but the large accuracy jump is at Phase 2, when we choose better crops. Early errors come more from a bad crop than from raw IoU alone. EfficientNet and Fusion-EN keep climbing in Phase 3. CLIP and SigLIP linear

TABLE II

END-TO-END FINAL-PROBABILITY TOP-1 (%) BY PIPELINE STAGE. DET.: MASK IOU. GROUP ROW: ANY-FRAME ANY-BBOX OVER 463 CLIPS WITH SPEECH-ALIGNED RERANKING.

Stage	CLIP	SigLIP	EffNet	Fusion	Fus.-EN
<i>Detection (mask IoU only)</i>					
Hand crop			Mask IoU = 8.3		
Full image			Mask IoU = 19.8		
Ph.1 Backbone	15.7	14.4	21.8	19.7	22.0
Ph.2 Mask select	38.8	36.9	49.9	46.4	51.3
Ph.3 Rerank pool	40.8	39.8	57.4	49.8	58.7
Ph.4 Margin rerank	39.9	39.0	57.5	49.7	58.6
Group (speech-align.)	56.8	55.1	69.6	69.1	73.4

TABLE III

SPEECH-ONLY CLIP ACCURACY (%) AND WER VS. THE COMMON-PHRASE TRANSCRIPT (463 CLIPS).

ASR	WER	Rules acc.	Semantic acc.
Whisper (no LoRA)	142.2	61.3	67.0
Whisper + LoRA	33.3	79.1	81.9

probes stay below 41% on frames even after reranking. Phase 4 barely changes frame scores, but it lets us add a Whisper-agreement bonus at clip-level evaluation. On clips, Fusion-EN reaches **73.4%**, ahead of EfficientNet (69.6%) and Fusion-CLIP (69.1%).

B. Speech Ablations

Table III scores the speech branch alone on the same 463 clips (no mask gate). GT for WER is the clip’s common phrase. Base Whisper often inserts extra words, so mean WER can exceed 100%; LoRA cuts it from 142.2% to 33.3%. MiniLM (semantic) beats text rules at both settings (67.0% vs. 61.3% without LoRA; 81.9% vs. 79.1% with LoRA). Speech-only Top-1 (LoRA+semantic, 81.9%) is higher than Fusion-EN (73.4%) because the end-to-end score still requires a valid mask; fusion helps most when vision is already close and the phrase names deposit vs. non-deposit.

C. Best Models and Confusion Analysis

On **frame-level** evaluation with the final pipeline, EfficientNet reaches 67.8% and Fusion-EN 70.0% final-probability Top-1 (3,197 frames with a working mask). Under the **any-frame** clip rule, Fusion-EN names the right class in 79.5% of the 463 clips before the mask gate, which the gate turns into the 73.4% reported in Table II. Fig. 3 compares the two frame-level matrices with the any-frame clip matrix of Fusion-EN.

Uneven class counts. The class distribution is highly imbalanced. *Other #1 Bottles* is 732/3,538 instances (20.7%). Garbage Bags (9), Other Plastic Film (22), Plastic (20), and Other Rigid/Bulky Plastics (18) are each <1% of labels. With even sampling, the head prefers the common class. On frames, 19.7% of Unnumbered containers, 17.3% of #2 Colored, and 17.0% of Other Plastics are called Other #1, even when the true resin is different. Rare classes have zero recall for Garbage Bags and Film with EfficientNet and stay below 35% for generic Plastic and Film even with Fusion-EN.

Mix-ups among similar classes. Five common misclassification groups show up: (1) **#1 bottle family**—Deposit #1, #1 Clamshells, and Other #1 share similar round PET geometries; (2) **#2 HDPE family**—#2 Natural vs. #2 Colored differ primarily in color, making them difficult to distinguish visually; (3) **Black/dark cluster**—Black Plastic takes errors from Deposit #1, Clamshells, and Unnumbered pieces when the hand covers the object; (4) **Catch-all vs. specific**—Unnumbered fragments and Other Plastics leak into numbered container classes; (5) **Rare/ambiguous materials**—Film, garbage bags, and generic Plastic have no rigid shape. Speech fusion improves performance when the associated verbal phrases are lexically distinct (e.g., deposit vs. non-deposit). However, its benefit is limited in cases where the segmentation mask fails to capture the target object, or where automatic speech recognition produces a truncated or corrupted transcript.

Fixing the uneven classes. Given that the Other-#1 hub accounts for a substantial proportion of classification errors, we retrained the image model with *inverse-frequency class weights*, where errors on rare classes would result in a higher penalty than errors on common classes. This cut the frame-level false-positive rate into Other #1 from 16.3% to 11.7% and raised macro-F1 by 1.2%, while clip accuracy stayed about the same (73.4% → 72.8%). Unnumbered container recall showed the largest improvement (+12.8%), consistent with expectations, as these items were the primary source of incorrect Other-#1 predictions. We additionally evaluated oversampling of rare-class images across training epochs; however, this approach did not yield improvement. As rare classes contain only 9–22 unique crops, oversampling caused the model to memorize specific instances rather than generalize to the underlying class. These results indicate that loss rebalancing helps visually similar container classes, whereas rare materials need additional data.

D. Qualitative Examples

Successes usually have a centered mask, a medium-size object, and a spoken phrase that sits in our list; failures often stack a missed hand crop, a chopped transcript (“5,” for #5 PP), or clear vs. colored #2 bottles in bad light. Late fusion recovers several deposit vs. non-deposit clips that EfficientNet misses when the bottle label is covered, but when both the mask and speech-to-text fail, no fusion model beats chance—later gains likely need better masks and audio together, not only a rerank after the fact.

VI. DISCUSSION

Vision-speech fusion provides the greatest benefit when mask quality is already high: Fusion-EN outperforms EfficientNet by 0.2% at Phase 1, but by 3.8% at clip-level evaluation, where audio tells deposit status and resin hints apart. Whisper plus phrase search fails when the transcript is empty or matches the wrong phrase; text-based rules mitigate but do not eliminate these failure cases. Class weighting is an inexpensive fix for uneven class counts: it trims the Other-#1 false-positive hub and lifts macro-F1 without extra data, but it does not help

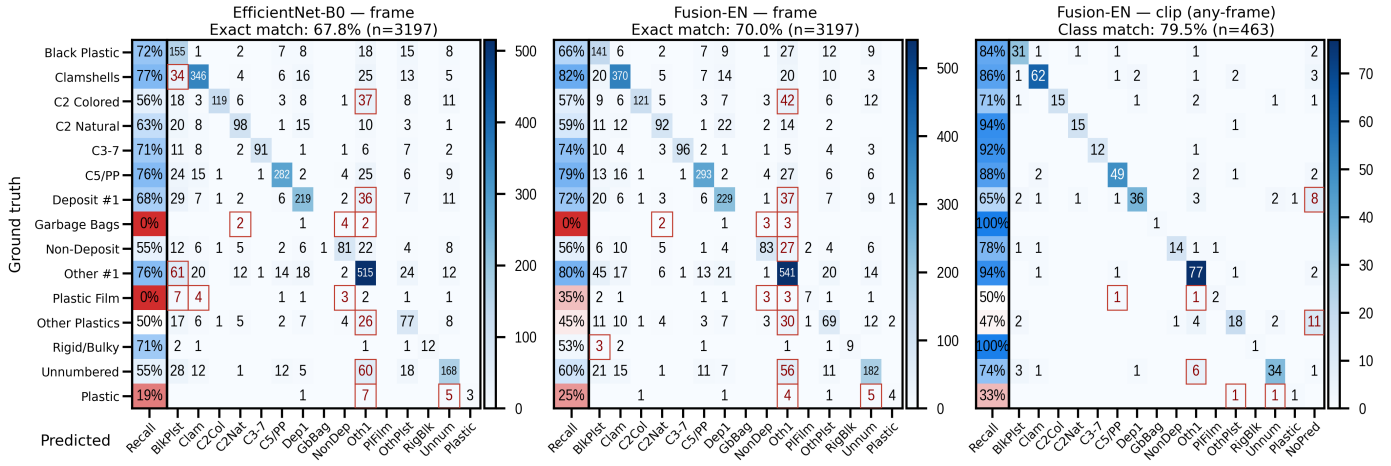


Fig. 3. Confusion matrices for the final pipeline: EfficientNet-B0 and Fusion-EN at the frame level, and Fusion-EN over the 463 clips under the any-frame rule used in Table II (clip correct when any frame predicts the labeled class; missed clips are charted with their majority prediction). The leftmost column of each panel is per-class recall; red boxes mark confusions above 7% of a row (15% for rows with few samples).

the rarest materials, and oversampling likewise fails without more examples. Limitations include a plastics-only 15-class taxonomy, hand-crop sensitivity, scarce rare-class data, and no glove-versus-object mask in the final pipeline. Separate #1/#2 heads, more data, and stronger detectors would likely help.

VII. CONCLUSION

We built a vision-plus-speech pipeline for detailed plastic sorting on EgoWaste. It uses SAM3 [14] masks, four-stage reranking, image classifiers, Whisper LoRA transcripts, phrase matching, and logistic regression at the end. Stage-by-stage scores show that masks and candidate reranking drive early gains, while speech adds a steady lift at the clip level (73.4% group any-frame accuracy for Fusion-EN).

REFERENCES

- [1] S. P. Gundupalli, S. Hait, and A. Thakur, "A review on automated sorting of source-separated municipal solid waste for recycling," *Waste Management*, vol. 60, pp. 56–74, 2017.
- [2] O. Tamin, E. G. Mounq, and A. Farzamnia, "Object detection for waste management: A comparative review of models, challenges, and future directions," *Bulletin of Electrical Engineering and Informatics*, vol. 15, no. 3, pp. 2191–2211, 2026.
- [3] M. Yang and G. Thung, "Classification of trash for recyclability status," Stanford CS229 Project Report, 2016.
- [4] D. Bashkurova, M. Abdelfattah, Z. Zhu *et al.*, "ZeroWaste dataset: Towards deformable object segmentation in cluttered scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2022, pp. 21 147–21 157.
- [5] D. Yudin, N. Zakharenko, A. Smetanin *et al.*, "Hierarchical waste detection with weakly supervised segmentation in images from recycling plants," *Eng. Appl. Artif. Intell.*, vol. 128, p. 107542, 2024.
- [6] K. Friedrich, G. Koinig, R. Pomberger, and D. Vollprecht, "Qualitative analysis of post-consumer and post-industrial waste via near-infrared, visual and induction identification with experimental sensor-based sorting setup," *MethodsX*, vol. 9, p. 101686, 2022.
- [7] H. Xu, W. Tang, Z. Li, K. Qin, and J. Zou, "Multimodal dual cross-attention fusion strategy for autonomous garbage classification system," *IEEE Trans. Ind. Inform.*, vol. 20, no. 11, pp. 13 319–13 329, 2024.
- [8] Arkansas Department of Energy and Environment, "Class 1 license — material recycling facility," Arkansas Code of Rules, 8 CAR §62-302, 2025.
- [9] Connecticut Department of Energy and Environmental Protection, "General permit to construct and operate a commercial facility for the management of recyclable materials and certain solid wastes," DEEP-MM-GP-001, 2022.

- [10] Environment Agency, United Kingdom, "SR2022 no. 7: Materials recycling facility," 2022.
- [11] Health and Safety Executive, United Kingdom, "Sorting and processing waste," HSE guidance, 2024.
- [12] K. Grauman *et al.*, "Ego4D: Around the world in 3,000 hours of egocentric video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2022.
- [13] V. Chavan, Y. Imgrund, T. Dao, S. Bai, B. Wang, Z. Lu, O. Heimann, and J. Krüger, "IndEgo: A dataset of industrial scenarios and collaborative work for egocentric assistants," in *Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets and Benchmarks Track*, 2025.
- [14] N. Carion, L. Gustafson, Y.-T. Hu *et al.*, "SAM 3: Segment anything with concepts," *arXiv preprint arXiv:2511.16719*, 2025.
- [15] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021.
- [16] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.
- [17] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019.
- [18] A. Radford, J. W. Kim, T. Xu *et al.*, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023.
- [19] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [20] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2019.
- [21] P. F. Proença and P. Simões, "TACO: Trash annotations in context for litter detection," *arXiv preprint arXiv:2003.06975*, 2020.
- [22] Z. Ma, Z. You, Y. Dong *et al.*, "UrbanWaste: In-the-bin dataset for waste disposal inspection with multi-granularity hierarchical labels," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 28 231–28 239.
- [23] S. Liu *et al.*, "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.
- [24] G. Radevski, D. Grujicic, M. Blaschko, M.-F. Moens, and T. Tuytelaars, "Multimodal distillation for egocentric action recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 5213–5224.
- [25] J.-B. Alayrac *et al.*, "Flamingo: a visual language model for few-shot learning," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [26] G. Lu, Y. Wang, H. Xu, H. Yang, and J. Zou, "Deep multimodal learning for municipal solid waste sorting," *Sci. China Technol. Sci.*, vol. 65, pp. 324–335, 2022.